CAR-GUIDE: Cost-Aware Reliability with Capability Evidence and Selective Inference

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Abstract

Tool-using agents operate under unequal uncertainty: some requested actions are unavailable, while mistakes in consequential state changes matter more than mistakes in read-only reasoning. We introduce CAR-GUIDE, an automotive assistant that combines deterministic capability evidence with semantic gates and consequence-selective inference. On CAR-bench, CAR-GUIDE achieves 75.8% official macro Pass^3, 17.3 observed points above a reproduced reference-agent configuration. We perform systematic ablation studies of seven mechanism groups. Removing the capability-aware path, selective self-consistency, advisory gates, or unavailable-result handling lowers observed macro Pass^3 by 5.1–7.1 points. We also compare selective, single-sample, and blanket-sampling policies. Selective sampling uses 50.5% fewer primary samples than voting at every observed decision and is substantially cheaper, without a resolved quality difference. These results support allocating reliability controls according to available evidence and decision consequence.

1 Introduction

Tool-using language-model agents must do more than complete a task once: they must repeatedly choose correct actions, ask for missing information, respect policies, and decline requests that exceed their capabilities. CAR-bench targets these properties in a multi-turn automotive assistant with mutable state, domain policies, and 57 application tools exposed to the agent [Kirmayr *et al.*, 2026]. Its Base tasks test ordinary completion, Hallucination tasks test whether agents appropriately handle missing tools or information without claiming unsupported success, and Disambiguation tasks require clarification or information gathering. The primary official score macro-averages task-family Pass^3, which requires all three trials of a task to pass and therefore amplifies inconsistent behavior.

We introduce CAR-GUIDE (Capability-Aware Reliability through Gated Use of Inference and Deterministic Evidence), which allocates controls by what is knowable and consequential. It compares visible tools with a fixed application contract, uses narrow gates for semantic ambiguity, and applies three-sample self-consistency only to proposed state changes.

Our contributions are as follows:

1. CAR-GUIDE, a reliable tool-using agent that reaches 75.8% official macro Pass^3 on CAR-bench;
2. a systematic, replicated ablation study identifying which controls contribute to reliability and on which task families (RQ1); and
3. a cost-aware study of selective versus uniform inference allocation (RQ2), the reference agent, and a higher-cost model tier (RQ3).

2 Related Work

ReAct established interleaved reasoning and environment action for tool-using agents [Yao *et al.*, 2023], and ToolLLM broadened API selection and use [Qin *et al.*, 2024]. Yet ToolBeHonest and FAIL-TaLMs show that agents still struggle to recognize requests that cannot be completed with the tools or information available to them [Zhang *et al.*, 2024; Treviño *et al.*, 2025]. Together, this work motivates making capability limits explicit rather than leaving them to model recall.

Self-consistency aggregates multiple answer samples [Wang *et al.*, 2022], and agent-specific variants aggregate trajectories or actions [Wang *et al.*, 2024]. This provides reliability through repeated inference, but uniform sampling spends the same compute on decisions with different consequences. Adaptive test-time computation instead allocates effort by estimated difficulty [Snell *et al.*, 2025; Damani *et al.*, 2025]; our setting allocates it by action consequence within an agent trajectory, concentrating additional samples on proposed state changes.

τ-bench introduced Pass^k to measure success across repeated agent executions [Yao *et al.*, 2025]. CAR-bench adopts this strict reliability view alongside retry-oriented Pass@k and adds tasks that test capability limits and clarification [Kirmayr *et al.*, 2026]. We use these metrics to separate reliable repeated success from recoverability across retries.

3 CAR-GUIDE Agent

The agent follows the reason–act interaction pattern common to tool-using agents [Yao *et al.*, 2023], but allocates controls

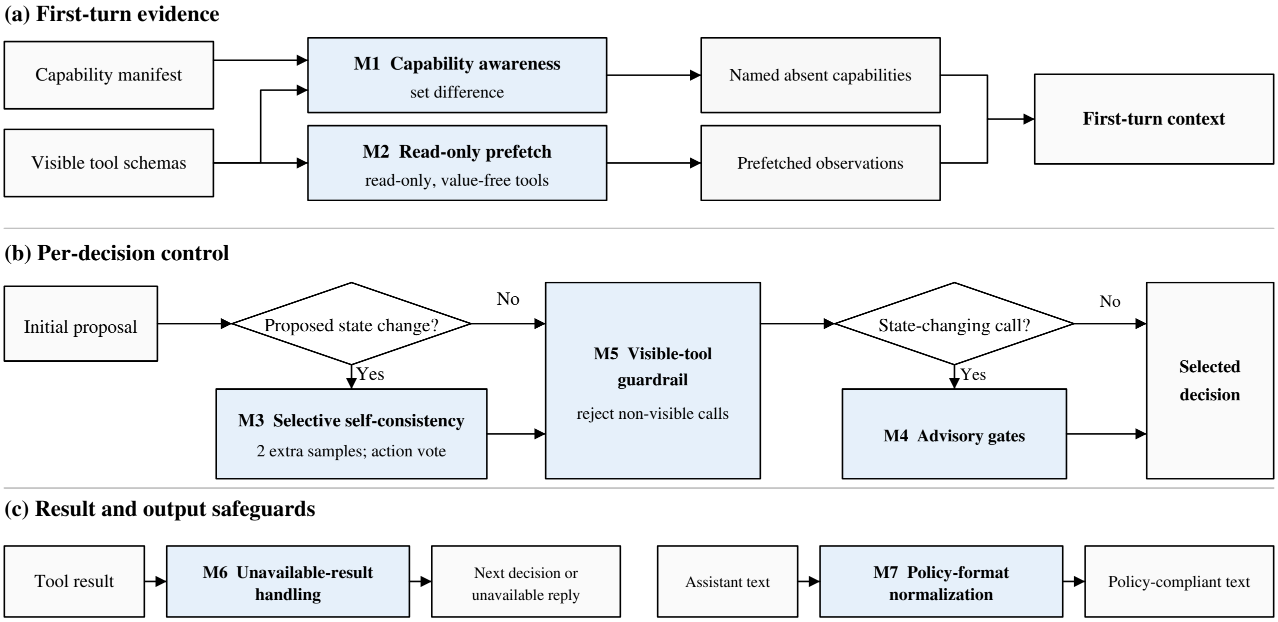

Figure 1: CAR-GUIDE's seven mechanisms operate at three scopes. M1–M2 construct first-turn evidence. Per decision, M3 repeats inference only for a proposed state change; M5 rejects non-visible calls, and M4 checks selected state-changing calls. M6 handles unavailable result evidence, including explicit `unknown` sentinels, before the next decision, while M7 normalizes policy-constrained output formatting. Blue marks CAR-GUIDE mechanisms; neutral boxes are shared inputs, processing, and outputs.

by a common rule: prefer deterministic evidence, invoke semantic judgments narrowly, and repeat inference only before consequential state changes (Figure 1). Its seven groups span deterministic checks, cached and uncached model judgments, inference allocation, and policy-specific normalization.

M1: Capability awareness. CAR-bench permits external knowledge bases and precomputed lookup tables that do not encode test-set answers [CAR-bench Challenge Organizers, 2026]. We therefore define a fixed manifest $\mathcal{M}$ of all application tool names. On the first turn, the agent compares it with the visible names $\mathcal{V}$ and injects the deterministic difference $\mathcal{M} \setminus \mathcal{V}$, instructing the model not to simulate missing operations or claim unsupported success. The manifest contains application metadata, not task answers.

M2: Read-only prefetch. On the first turn, a model classifies structurally value-free tool schemas as read-only; those selected are prefetched to expose current state before the agent decides. Restricting prefetch to value-free schemas avoids inventing arguments.

M3: Selective self-consistency. On the first turn, the model classifies each visible tool as read-only or state-changing, and the agent caches this mapping for later decisions. Read-only calls, clarifications, and terminal responses use one initial sample; proposed mutations draw two additional samples concurrently and execute the plurality action signature. Ties retain the initial sample.

M4: Narrow advisory gates. Before executing selected state changes, two model-mediated gates reconsider the proposed calls. When capabilities are absent, a substitution gate removes workarounds that would misrepresent success; a necessity gate removes extraneous mutations.

M5: Visible-tool guardrail. The system prompt names the currently visible tools and forbids calls to hidden or remembered names. After generation, a deterministic check replaces non-visible calls with a response stating that the agent cannot perform the action.

M6: Unavailable-result handling. Prompt rules instruct the model to treat absent result fields as unavailable rather than false or default values. In addition, when a successful structured tool result explicitly marks a top-level field as `unknown`, a deterministic check returns a response stating that the required evidence is unavailable.

M7: Policy-format normalization. When the active policy requires 24-hour time, a deterministic postprocessor normalizes 12-hour expressions in assistant text and in the natural-language email argument.

4 Evaluation

Protocol. We hold out the 125 public-test tasks for final agent selection and conduct all reported analyses on the 129 public-training tasks (50 Base, 48 Hallucination, and 31 Disambiguation). Each complete batch contains three stochastic trials per task under the pinned CAR-bench harness. A lightweight model and reasoning-effort sweep on a stratified 36-task subset selected DeepSeek-V4 Flash and Pro [DeepSeek-AI, 2026] at xhigh reasoning for evaluation.

Official macro Pass^3, the unweighted task-family average of success in all three executions, is the primary metric [Kirmayr

Configuration	Overall macro Pass^3 (%)	Base	Hallucination	Disambiguation
Full CAR-GUIDE	75.8 [75.3,76.3]	78.0 [76.0,80.0]	88.2 [83.3,91.7]	61.3 [58.1,64.5]
Reproduced CAR-bench reference	58.5 [56.7,60.6]	72.0 [70.0,74.0]	58.3 [56.2,62.5]	45.2 [41.9,48.4]
M1 path off: capability awareness + gate	70.5 [67.7,72.1]	78.0 [76.0,80.0]	70.1 [62.5,75.0]	63.4 [61.3,64.5]
M2 off: read-only prefetch	73.8 [71.4,75.1]	76.7 [72.0,80.0]	93.1 [91.7,93.8]	51.6 [48.4,54.8]
M3 off: selective self-consistency	70.7 [68.5,72.7]	69.3 [62.0,80.0]	86.8 [83.3,91.7]	55.9 [54.8,58.1]
M4 off: advisory gates	68.7 [65.6,70.5]	68.7 [68.0,70.0]	82.6 [77.1,85.4]	54.8 [51.6,58.1]
M5 off: visible-tool guardrail	78.0 [74.3,79.9]	77.3 [74.0,82.0]	91.0 [87.5,95.8]	65.6 [61.3,67.7]
M6 off: unavailable-result handling	69.1 [65.9,72.9]	72.7 [68.0,76.0]	74.3 [70.8,77.1]	60.2 [54.8,67.7]
M7 off: policy-format normalization	73.8 [71.7,77.1]	76.7 [76.0,78.0]	86.8 [85.4,87.5]	58.1 [51.6,67.7]

Table 1: Reference comparison and task-family profiles of the seven prespecified mechanism-group removals. Every entry is a three-batch mean [observed range] in percent.

Task family	Tasks	Trials	Primary demand
Base	50	150	complete supported requests
Hallucination	48	144	detect missing capability/info.
Disambiguation	31	93	clarify or gather information
Total	129	387	

Table 2: Composition of every admitted full-pool quality batch. The separate latency microbenchmark uses 12 tasks and 36 trials per policy.

et al., 2026]. Agent cost excludes simulation and judging.

Controlled comparisons. We remove one prespecified mechanism group at a time: M1-off disables the capability-aware path and its dependent substitution gate; M2-off disables first-turn prefetch; M3-off uses one sample; M4-off bypasses both advisory gates; M5-off removes visible-tool prompt discipline and rejection; M6-off removes unavailable-result guidance and sentinel handling; and M7-off disables 24-hour normalization. Blanket $K = 3$ instead applies voting to every primary decision. Every arm has three complete 387-trial batches. Code, configurations, per-trial results, metric definitions, and run provenance are provided in the release artifact.

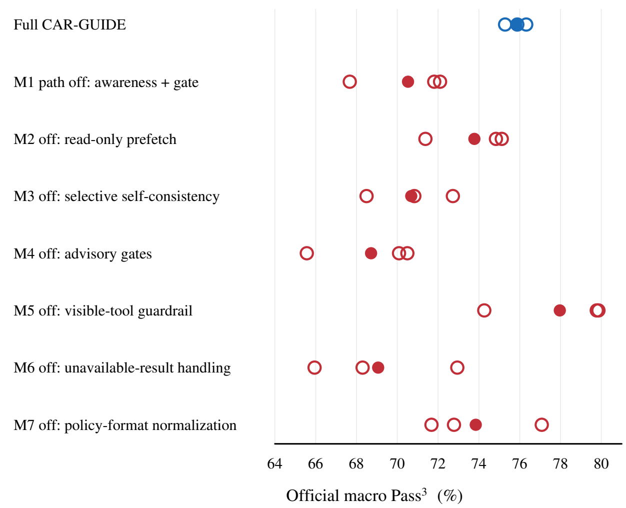

Figure 2: Batch-level replication of the headline ablations on the primary official macro Pass^3 metric. Hollow circles are complete 129-task stochastic batches and filled circles are arm means. Every treatment batch falls below every full-system batch for the M1 path, M3, M4, and M6. M2 has only a 0.16-point gap between its best batch and the weakest control, while M5 and M7 overlap the full-system range.

5 Results and Discussion

CAR-GUIDE averages 75.8% official macro Pass^3 over three complete batches, compared with 58.5% for the reproduced CAR-bench reference configuration using the same DeepSeek-V4 Flash model and xhigh reasoning effort. We perform structured ablations to identify where this improvement comes from.

RQ1: Which controls matter, and where? Removing the composite capability-aware path, M3, joint M4, or M6 is associated with mean official macro Pass^3 reductions of 5.3, 5.1, 7.1, and 6.8 points, respectively, with every removal batch below every full-system batch (Figure 2). Exploratory family profiles differ (Table 1): the capability-path difference is largest on Hallucination, M3 differs most on Base and Disambiguation, M4 is lower across all three families, and M6 is lower primarily on Hallucination (its matched Base mean is slightly higher). M2-off is only -2.0 points overall but combines a -9.7-point Disambiguation shift with $+4.9$ on Hallucination, showing opposing observed family shifts. M5-off overlaps control and has a higher mean; M5 is therefore retained for deterministic schema integrity, not a resolved mean benefit. M7-off is -2.0 points with an overlapping range, so its aggregate benefit is unresolved. However, direct output auditing finds that M7 eliminates all observed 12-hour-format violations: 0/237 time-bearing output units violate the policy with M7, compared with 52/240 (21.7%) without it.

RQ2: Does selective voting improve the measured allocation? We compare three inference-allocation policies. $K{=}1$ uses only the initial sample at every decision; selective $K{=}3$ draws two additional samples only when that sample proposes a state change; and blanket $K{=}3$ draws three samples at every primary decision. $K{=}1$ averages 70.7% macro Pass^3 versus 75.8% for selective $K = 3$. Tables 3 and 4 report complementary single-batch reliability profiles. Blanket $K = 3$ averages a similar 74.2% but costs 53.3% more than selective voting (Figure 3). Relative to $K = 1$, selective voting gains 5.1 observed macro Pass^3 points for \$0.00135 more per trial.

Policy	Pass^1	Pass^2	Pass^3	Pass@1	Pass@2	Pass@3	Gap@3
$K = 1$	81.7%	74.9%	69.8%	81.7%	88.4%	89.9%	20.2%
Selective $K = 3$	84.8%	79.8%	76.7%	84.8%	89.7%	91.5%	14.7%
Blanket $K = 3$	85.5%	81.4%	79.8%	85.5%	89.7%	92.2%	12.4%

Table 3: Repeated-execution reliability and retry potential on the pooled 129-task public training set. Pass^k estimates success in all k sampled executions; Pass@k estimates at least one success. Gap@3 is Pass@3 minus Pass^3.

Policy	0/3	1/3	2/3	3/3	Pair agree
$K = 1$	13	6	20	90	86.6%
Selective $K = 3$	11	7	12	99	90.2%
Blanket $K = 3$	10	10	6	103	91.7%

Table 4: Distribution of successful executions per task. Pair agree is correctness agreement for a uniformly selected pair of the three trials; it counts consistent failure as agreement and is interpreted only alongside success.

A canonical audit finds that only 390/1,606 primary decisions (24.3%) trigger voting. It draws 50.5% fewer primary samples than counterfactual voting at every decision observed in that audit and changes the initial action 29 times. Audit and ablation batches are independent, so these counts establish selective operation rather than per-intervention causality.

RQ3: How does CAR-GUIDE compare with the reference agent and a higher-cost model? The CAR-bench reference-agent configuration averages 58.5% macro Pass^3 at \$0.00485 per trial, whereas CAR-GUIDE averages 75.8% at \$0.00766. Thus the complete agent gains 17.3 points for an additional \$0.00281 per trial on the public training pool.

Exact-source Flash and Pro average 100.3 and 101.0 of 129 tasks passing all three trials, but Pro costs 5.87× as much. Their observed ranges overlap, supporting only a measured operating-point comparison.

The overlapping selective and blanket ranges do not establish equivalence; they support only the measured cost difference without a resolved quality difference.

Across the three questions, the experiments support allocating controls by task demand, available evidence, and decision consequence rather than applying them uniformly. Mutations persist while reads and dialogue do not, motivating extra inference at the former. Likewise, a fixed manifest distinguishes unsupported operations from temporarily absent ones. The capability-path difference is largest on Hallucination and matches that path's intended scope.

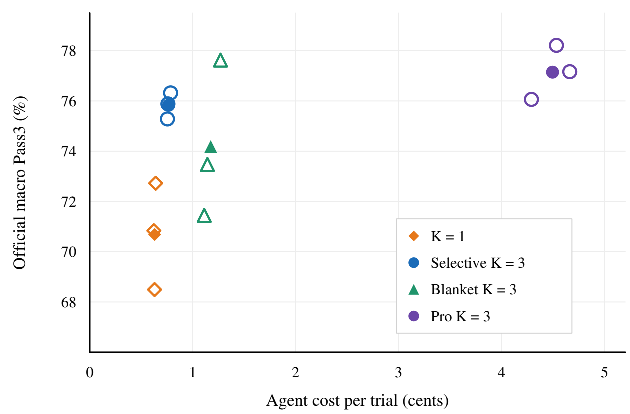

Figure 3: Observed quality–cost operating points. Hollow markers are complete stochastic batches and matching filled markers are arm means; every arm has three batches. The submitted selective policy has higher observed reliability than $K = 1$ at modest additional cost, whereas blanket voting is substantially more expensive without a resolved quality advantage. Pro is over five times as costly as submitted Flash with overlapping observed quality. Lines are intentionally omitted: the data do not establish a Pareto frontier.

6 Limitations and Conclusion

Three batches, a development-visible training pool, and overlapping mechanism groups limit causal and generalization claims; Pass^3 also ignores severity. Within these bounds, we introduced CAR-GUIDE, demonstrated 75.8% official macro Pass^3 on CAR-bench, and systematically studied its reliability and cost. Replicated ablations associate the capability-aware path, selective self-consistency, advisory gates, and unavailable-result handling with repeated success. Selective inference concentrates additional computation on state-changing decisions, improving observed reliability over single-sample inference while avoiding the cost of blanket voting. CAR-GUIDE therefore offers a reproducible example of matching agent safeguards and computation to the evidence available and the consequence of each decision.

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